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Q1

Binary search is $O(n)$ in the worst case because if the element is missing you may have to look at every cell to be sure. The recurrence just shows the best case is faster.

B11902003

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Q2

Exchange argument: let $G = g_1..g_k$ be greedy's picks (by finish time) and $OPT = o_1..o_m$ an optimal solution, both sorted. By induction $f(g_i) \leq f(o_i)$: greedy picks the earliest-finishing compatible interval, so we can swap o_i for g_i without breaking compatibility. Hence $k \geq m$, so greedy is optimal.

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Q3

Let $L[i]$ = length of the LIS ending exactly at $a[i]$.

$L[i] = 1 + \max\{ L[j] : j < i, a[j] < a[i] \}$ (or 1 if none). Answer $\max L[i]$.

Two nested loops give $O(n^2)$. Refinement: keep $tails[k]$ = smallest tail of an increasing subsequence of length $k+1$; binary-search each element: $O(n \log n)$.

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Q4

True. Invariant: when BFS finishes layer d , every vertex at distance d has been discovered with $\text{dist} = d$. Base: layer 0 is $\{s\}$. Step: any v at distance $d+1$ has a neighbour u at distance d ; when u is dequeued v is discovered with $\text{dist } d+1$, and no earlier layer can reach v or its distance would be $< d+1$.